

BDH-Inspired Recurrent Reasoning Lab

Technical Report and Explorable Learning Artifact

Submission scope: Educational/research-scale implementation inspired by publicly documented BDH and BDH-CQ concepts. This is not a reproduction of Pathway's proprietary production system.

Abstract

This project presents an interactive, research-scale laboratory for studying sparse recurrent computation, associative memory, latent iterative reasoning, interpretability telemetry, and structured transformation reasoning. The system combines a React interactive playground with a FastAPI backend, generated ARC-style tasks, recurrent latent computation, associative memory, a transformation hypothesis engine, quantitative telemetry, and reproducible controlled evaluation. A corrected controlled benchmark containing 360 episodes achieved 360/360 exact matches. This result is deliberately scoped to the project's generated benchmark and is not presented as ARC-AGI, BDH-CQ, or Pathway performance.

1. Motivation

Many modern language models rely heavily on attention-based sequence processing. Recent research also explores recurrent, state-based, and memory-oriented approaches that can process information through persistent internal state.

BDH introduces a biologically inspired architecture involving locally interacting neurons, sparse positive activations, recurrent computation, and synaptic-plasticity-based working memory. BDH-CQ extends this direction toward in-context learning with recurrent latent reasoning. These ideas motivate a practical question: can a smaller, transparent implementation provide an experimental environment in which recurrent memory and latent reasoning can be manipulated, observed, and stress-tested?

2. Research Questions

Primary: Can a sparse recurrent architecture with associative memory and latent iterative computation learn and apply structured transformations from demonstrations while remaining computationally inspectable and avoiding the need for verbalized chain-of-thought?

Secondary: Where does such an architecture fail as task complexity increases, and can those failure modes be detected or mitigated?

3. Relationship to Public Research

Dragon Hatchling (BDH). The BDH paper describes a scale-free, biologically inspired network of locally interacting neuron particles and discusses sparse positive activations, recurrent computation, synaptic plasticity, and interpretability.

BDH-CQ. BDH-CQ investigates in-context learning with recurrent latent reasoning. Its published benchmark results are not the results of this project.

Related research. Hungry Hungry Hippos provides broader context for state-space sequence modeling, while Titans studies neural memory and test-time memorization.

4. Project Architecture

React Playground → FastAPI Backend → Task Generator → Model Engine → Transformation/Hypothesis Engine → Evaluation and Telemetry → Interpretability and Failure Analysis.

5. Demonstration-Conditioned Memory

A simplified experimental memory update is $S_t = (1-\gamma)S_{t-1} + \eta(A_t B_t^T)$. The purpose is to study how information from previous demonstrations can influence later computation through persistent state. This is an experimental implementation inspired by broader memory concepts and is not an exact reproduction of Pathway's internal memory implementation.

6. Recurrent Latent Reasoning

After demonstrations are processed, the query is encoded and passed through a recurrent latent workspace. A simplified abstraction is $H_{r+1} = F_\theta(H_r, S_K)$. The playground allows reasoning depth to be changed experimentally.

7. Sparse Computation and Telemetry

The experimental model includes a sparse neuron layer using positive nonlinear activations. Diagnostics include activation sparsity, memory norm, effective rank, entropy, interference-related metrics, reconstruction error, reasoning steps, and halting step. These are diagnostic measurements, not a human-readable chain-of-thought.

8. Transformation Hypothesis Engine

Candidate transformations are tested against demonstrations, inconsistent candidates are rejected, remaining candidates are ranked, and the selected transformation is applied to the query. Current families include translation, color transformation, and boundary propagation. This is a project-level experimental mechanism, not an official BDH-CQ component.

9. Interactive Task Families

Boundary Propagation, Color Transformation, and Translation are currently included. The infrastructure is designed for additional task families.

10. Interactive Learning Artifact

The browser playground allows users to change task family, memory mode, reasoning depth, number of demonstrations, grid size, random seed, adaptive halting, and memory rereading. The learner can change a computational assumption and immediately observe a prediction, evaluation metrics, transformation selection, and telemetry.

11. Controlled Evaluation

Configuration: 3 task families; 2 memory modes; 3 reasoning depths; 20 seeds; 5×5 grids; 3 demonstrations; 360 total episodes.

12. Verified Result

360 episodes; 360 exact matches; 100% exact-match rate; 100% mean accuracy.

This establishes reliable performance on the defined controlled benchmark. It does not establish ARC-AGI, BDH-CQ, Pathway benchmark, human-level reasoning, or arbitrary generalization. The benchmark is saturated because all tested configurations reached the same result, so it does not show that deeper reasoning or a particular memory mode is superior.

13. Debugging and Benchmark Validation

Initial Boundary Propagation errors were traced to inconsistent corner semantics between the task generator and transformation implementation. After aligning the semantics, all 20 tested seeds achieved exact matches. This illustrates why benchmark generation and evaluator logic must be validated before attributing failures to the model.

14. Original Project Contributions

Interactive research playground; latent telemetry; transformation consistency verification; empirical failure-boundary discovery; interference-oriented diagnostics; reproducible seeded evaluation.

15. Limitations

Limited benchmark difficulty; benchmark saturation; limited external generalization evidence; research-scale model size; no access to Pathway proprietary implementation; telemetry is not a complete semantic explanation; failure-risk mechanisms require harder independent validation.

16. Future Work

Expand task families; introduce harder multi-step transformations; add held-out and OOD tasks; sweep reasoning depth; compare memory mechanisms and lightweight baselines; increase grid and demonstration complexity; evaluate external ARC datasets; improve failure-risk calibration; study compute and memory scaling.

17. Reproducibility

Seeds 1–20; grid 5×5; 3 demonstrations; reasoning depths 1, 4, 8; memory modes baseline and simple_hebbian. Results are stored in results/controlled_evaluation.csv.

18. Scientific Integrity

Proprietary Pathway results are not represented as reproduced results. Generated benchmark results are explicitly labeled. The 100% result is not presented as ARC-AGI performance. Diagnostic telemetry is not described as literal chain-of-thought. Failure thresholds are not presented as universal limits.

19. Conclusion

This project does not claim to reproduce BDH-CQ. Instead, it builds an interactive laboratory around publicly described principles of sparse computation, associative memory, latent recurrence, structured transformations, interpretability telemetry, and failure analysis. The current benchmark is solved perfectly; the next research step is to make the benchmark harder, introduce held-out generalization tests, and determine where the architecture succeeds, where it fails, and whether its telemetry helps identify those failure modes.

References

- Kosowski et al. (2025), *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*, arXiv:2509.26507 — <https://arxiv.org/abs/2509.26507>
- Engdahl et al. (2026), *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888 — <https://arxiv.org/abs/2608.09888>
- Behrouz, Zhong & Mirrokni (2024/2025), *Titans: Learning to Memorize at Test Time*, arXiv:2501.00663 — <https://arxiv.org/abs/2501.00663>
- Fu et al. (2022), *Hungry Hungry Hippos: Towards Language Modeling with State Space Models*, arXiv:2212.14052 — <https://arxiv.org/abs/2212.14052>

- Pathway, Kharagpur Data Science Hackathon 2026 — <https://pathway.com/events/kharagpur-data-science-hackathon-2026>
- Pathway BDH public repository — <https://github.com/pathwaycom/bdh>

AI Assistance Disclosure

AI tools were used during development for code-generation assistance, debugging, documentation drafting, architecture discussion, experiment-design discussion, and technical explanation. AI-assisted material was reviewed and integrated by the project team. Experimental results reported here correspond to executions of the project code and stored benchmark outputs.